Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/05/2012
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of Artificial Gravity in Promoting Tissue-Regenerative Matrix-Integrin-Kinase Cell Signaling (Ground/Flight)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/01/2009	**End Date:**	11/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 ; STS-135 NOTE: End date is now 11/30/2012 per A. Chu/ARC (Ed., 4/11/12) NOTE: Start/end dates changed to 6/1/09-5/31/12 from 5/1/09-4/30/12, per A. Chu/ARC (6/2009)		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The broad working hypothesis underlying the work we are reporting on here is that in space, the mechanical unloading caused by microgravity can affect the growth and regenerative health of a broad range of mammalian tissues. Conversely, we also hypothesized that normal gravity on earth activates a key, complex cell-signaling pathway responsible for maintaining normal regenerative tissue growth, the extracellular matrix(ECM)-integrin-kinase signaling pathway. The work we proposed was centered on the specific hypothesis that somatic stem cells responsible for tissue regeneration may require the mechanical stimulation of gravity to proliferate and differentiate, and to regenerate tissues at normal rates, and that microgravity may slow down tissue regenerative processes. Specifically we sought to investigate the role of gravity-generated forces in promoting tissue regeneration, and the molecular mechanisms by which gravity in stem cells is translated into proliferation and differentiation. During the course of the proposed work we had the opportunity to expand ground artificial gravity studies to microgravity during rapid turnaround spaceflight opportunities in the Space Tissue Loss (STL1) and (STL2) experiments with stem cell differentiation. This work was integrated in the ongoing grant research expanding the artificial gravity approach to the questions to also include microgravity. Specifically we aimed to broadly test the hypotheses that: 1) Gravity-dependent mechanical loading regulates cell proliferation and differentiation in the in-vivo newt tail regeneration model 2) Gravity-dependent mechanical-load signaling in somatic stem cell progenitors regulates their proliferation and differentiation in-vitro In addressing Aim 1, we determined that in the newt tail regeneration model, varying gravity mechanotransduction from microgravity to hypergravity does play an important role in regulating the proliferation and differentiation of somatic stem cell progenitors present in the newt tail blastema, determining the extent of tail regeneration and whether progenitor stem cells switch from an early proliferative state into a late, functional, differentiated state, validating our hypothesis for in-vivo tissue regeneration, and suggesting that to the extent that normal tissue regenerative processes in mammals are analogous to newt tail regeneration, that they too will likely be dependent on gravity mechanical loading to occur normally, and that microgravity will inhibit them. This conclusion offers a strong note of caution about long-term human exposure to microgravity, suggesting that organ systems and tissues dependent on somatic stem cell-based regeneration for normal health will likely fail to work normally, possibly resulting in significant tissue and organ degeneration. In addition our in-vitro studies with mouse embryonic stem cells also, and in particular the microgravity spaceflight component of this grant with the STL1 and STL2 experiments with mouse embryonic stem cell differentiation into embryoid bodies and keratinocytes, offered similar conclusions with added molecular mechanism details. Specifically we observed the inhibition of stem cell differentiation and failure of numerous molecular markers of tissue differentiation to occur in microgravity, and the persistence of stem cell markers that should normally switch off in response to initiation of differentiation plus large decreases in expression of cell adhesion and signaling molecules in the matrix-integrin-kinase axis. Overall the work we conducted using live animals, as well as cellular and molecular approaches consistently shows mechanical load is required for stem cells to differentiate and function in tissue regeneration, and that microgravity inhibits this process. Future work is needed to elucidate the specific molecular mechanisms linking gravity to tissue regeneration. Below we specifically report on Aim 1 under the title “Gravity Mechanical Load Modulates Tissue Growth and Regeneration in the Newt Pleurodeles waltl” on the work pertaining to the newt tail regeneration in-vivo model in hypergravity, and on results from the Foton M3 spaceflight experiment we analyzed in the context of this grant as part of obtaining results for tissue regeneration going from unloaded condition in aquaria, and microgravity, as well as under normal load at 1g and hypergravity at 2g. In total these results offer a very complete picture of responses to varying regenerative responses to increasing load, and of the importance and necessity of gravity for normal tissue regenerative processes. In addition we also report on Aim 2 results under the title “Microgravity Reduces the Proliferation, Differentiation, and Regenerative Potential of Mouse Embryonic Stem Cells” on the cell biology work pertaining to the in-vitro stem cell regeneration model based on induction of differentiation under mostly microgravity conditions, but also hypergravity. The cell biology results clearly show a gravity-dependent switch between stem cells and their functional differentiated descendents, and suggest microgravity is a strong inhibitor of regenerative processes. NOTE: This proposal has a companion spaceflight component entitled, "The Effects of Microgravity on the Tissue Regenerative Potential of Stem Cells--Flight." The spaceflight experiments are relevant to Aim 2. Having completed the initial data analysis of the STL2-SCR flight experiments conducted on STS-135, and for context, we also report here on the companion flight proposal progress and results. NOTE that the Flight and Ground portions of the project were combined in April 2011 for reporting purposes (Ed. 7/11/2011).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We believe the studies underway in this project are highly relevant to NASA because the hypothesis we are testing test, if validated, will be of key importance to understanding and maintenance of human tissue regenerative homeostasis and health during long-term spaceflight. The data we have already obtained in complementary spaceflight and ground studies clearly and reproducibly show that tissue regeneration in animal and cell models is significantly slower in space, and that gravity-generated forces are of key importance to stimulate the growth, proliferation, and differentiation of cells and tissues. In this study we asked the specific question of whether tissue regeneration in vivo, in the newt tail regeneration model, and regenerative somatic stem cell models, in vitro respond to hypergravity with increased regenerative proliferation, and if so by what molecular signaling mechanisms. These questions are of key importance because hypergravity in space is thought to be one of the few methods capable of compensating for the degenerative tissue conditions now know to be induced by long-term spaceflight. The results obtained in this grant thus far, with stem cell and newt tail regeneration experiments under hypergravity conditions, strongly support our initial hypothesis. The in-vivo newt tail regeneration in particular, for the first time shows vertebrate tissue regeneration being enhanced by gravity stimulation. In addition, companion spaceflight experiments in the STS-131 STL-SCR and STS-135-STL2-SCR, studying stem cell growth and differentiation, also point to microgravity as causing a failure of differentiation of tissue precursor cells in embryoid bodies and keratinocytes. In conclusion, this study shows promise of having great impact in

	this field by providing specific answers to the important question of whether and how gravity can promote regenerative proliferation and differentiation in relevant cell and animal models of regeneration. These novel insights into the potentially broad role of adult stem cell health in spaceflight-induced tissue degenerative conditions are likely to be an important factor in shaping the direction of future approaches to health during spaceflight, as well as being translated into medical advances in tissue regenerative health.
Task Progress:	During the last reporting period from 2011 to 2012 we focused our efforts on completing testing the hypothesis elaborated in Aim 2 of the grant proposal, that stem cells responsible for tissue regeneration have their proliferation and differentiation affected by gravity mechanostimulation. Specifically we had already tested in the previous year the differentiation of mouse embryonic stem cells into embryo-like cell massses (embryoid bodies) of various lineages and specific types and had found that microgravity inhibited the transition from stem cells to differentiated tissues, and that gravity mechanostimulation promoted it, as indicated both by gene expression and cell biology studies of differentiation markers and cell function. Those studies however, were not a of a specific cell type, so for the Space Tissue Loss 2 experiment flight opportunity on STS-135 we chose and proposed to study microgravity effects on keratinocyte differentiation from embryonic stem cells. The choice of keratinocyte differentiation as a model to study the effects of microgravity versus 1g on stem cell differentiation addresses the key question in Aim 2 of our proposal for a specific cell type, which was not possible with the mixed-lineage embryoid body model flown on STL1, and in addition addresses the important issue of wound closure by keratinocytes in microgravity. To do this, we used the time period between April and July 2011 to develop and ground verify the spaceflight protocol for differentiation of mouse embryonic stem cells into keratinocytes, and after this testing was completed we successfully flew the STL2 keratinocyte experiment on the final space shuttle mission, STS-135. Specifically, for the cell culture flight experiments, we used hollow fiber bioreactors seeded with stem cells and initiated differentiation immediately prior to flight with specific growth and transcription factors and cell culture medium to induce keratinocyte differentiation. After 15 days spaceflight and just prior to landing a set of flight bioreactors was fixed in space with RNA later to preserve mRNA, while another set was brought back to earth for live cell functional assays, such as cell migration in an vitro wound-healing assay. In the period following the flight we conducted ground based sample preparation qPCR analysis of gene expression and of post-flight cell cultures The analyses performed also included migration quantification on wound healing assays, and histocytochemistry. As a result of these studies we already determined that keratinocytes differentiated in microgravity with altered expression of keratins and reduced migratory ability, suggesting that differentiation and function of keratinocytes in wound closure would be diminished in microgravity. Finally, to complete the studies on stem cell proliferation and gene expression in response to gravity mechanostimulation we conducted cell culture centrifugation studies of mouse embryonic stem cells at 1g vs 25g. In these studies we observed no alterations in proliferation as we had previously observed in differentiated osteoprecursors, suggesting that stem cells and later stage progenitor lineages respond to mechanical stimulation in different ways. Our gene expression studies suggest that in stem cells gravity mechanostimulation affects differentiation and stemness markers while leaving proliferation regulators unchanged. We are currently preparing the results from these studies for publication.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
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